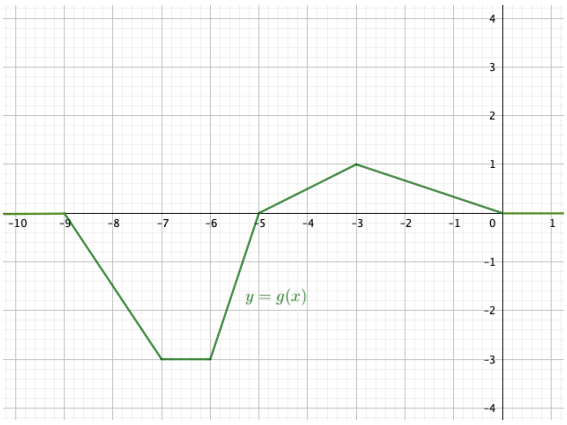
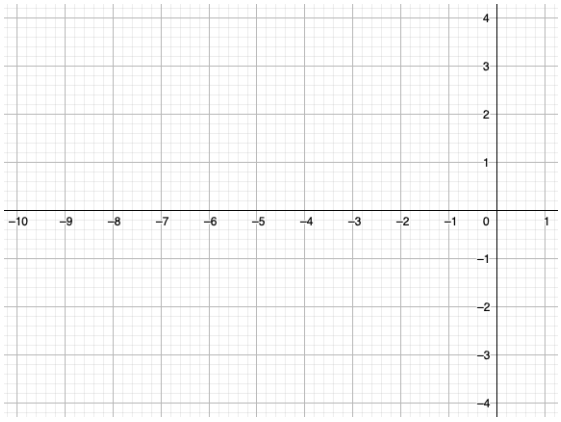
MATH 202 Your name: _____

Learning target I2, version 1

Given appropriate data about a function, determine outputs, direction, and concavity of its antiderivative, and/or construct an accurate graph.

Consider the piecewise linear function $y = g(x)$ pictured below; assume that $g(x) = 0$ for $x \leq -9$ and $x \geq 0$.





Let G be an antiderivative of g that satisfies $G(-9) = 3$.

1. On what intervals is G increasing? decreasing? neither? Carefully explain your answers.

2. On what intervals is G concave up? concave down? neither? Carefully explain your answers.

3. Recall we are given that $G(-9) = 3$. Find the exact values of $G(-7)$, $G(-6)$, $G(-5)$, $G(-3)$, and $G(0)$. Write one sentence to explain briefly how you found these values.

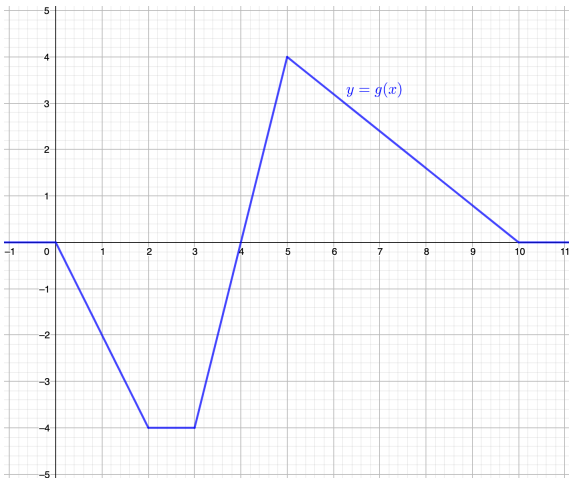
4. On the blank grid at right, sketch an accurate, labeled graph of $y = G(x)$.

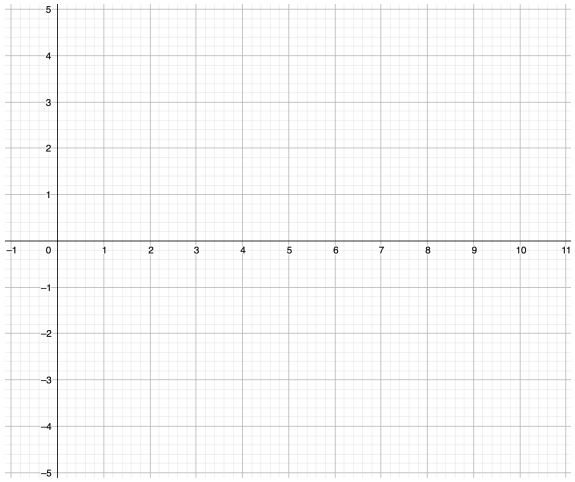
MATH 202 Your name: _____

Learning target I2, version 2

Given appropriate data about a function, determine outputs, direction, and concavity of its antiderivative, and/or construct an accurate graph.

Consider the piecewise linear function $y = g(x)$ pictured below.





Let G be an antiderivative of g that satisfies $G(0) = 3$.

1. On what intervals is G increasing? decreasing? neither? Carefully explain your answers.

2. On what intervals is G concave up? concave down? neither? Carefully explain your answers.

3. Find the exact values of $G(2)$, $G(3)$, $G(4)$, $G(5)$, and $G(10)$. Write one sentence to explain briefly how you found these values.

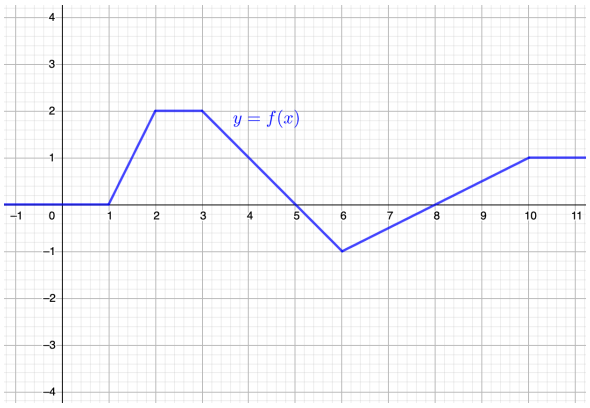
4. On the blank grid at right, sketch an accurate, labeled graph of $y = G(x)$.

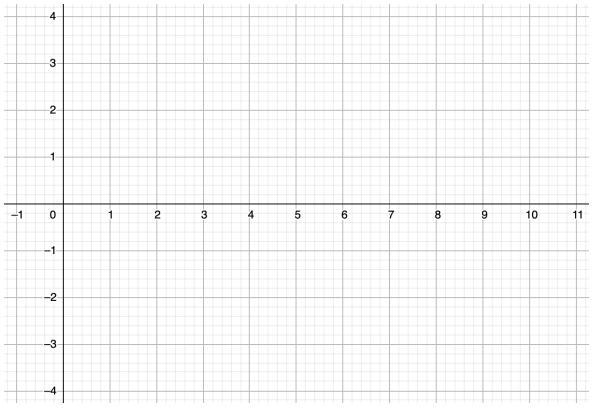
MATH 202 Your name: _____

Learning target I2, version 3

Given appropriate data about a function, determine outputs, direction, and concavity of its antiderivative, and/or construct an accurate graph.

Consider the piecewise linear function $y = f(x)$ pictured below; assume that $f(x) = 0$ for $x \leq -1$ and $f(x) = 1$ for $x \geq 11$.





Let F be an antiderivative of f that satisfies $F(1) = -2$.

1. On what intervals is F increasing? decreasing? neither? why?

2. On what intervals is F concave up? concave down? neither? why?

3. Find the exact values of $F(2)$, $F(3)$, $F(5)$, $F(6)$, $F(8)$, and $F(10)$. Write one sentence to explain briefly how you found these values.

4. On the blank grid at right, sketch an accurate, labeled graph of $y = F(x)$.